CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 10: Isosceles Triangles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In triangle ABC, $AB = AC$ and $\angle B = 52^\circ$. Find $\angle A$.
2. The vertical angle of an isosceles triangle is 100°. Find each base angle.
3. In triangle PQR, $\angle P = 50^\circ$ and $\angle R = 80^\circ$. Name the equal sides.
4. Each base angle of an isosceles triangle is 15° more than the vertical angle. Find the angles.
5. In triangle ABC, $AB = AC$ and BC is produced to D. If $\angle ACD = 130^\circ$, find $\angle A$.
6. An isosceles triangle has equal sides 10 cm and base 12 cm. Find the altitude to the base.
7. In triangle ABC, D is on BC with $AB = AD = DC$ and $\angle B = 64^\circ$. Find $\angle C$.
8. One angle of an isosceles triangle is 60°. What kind of triangle is it?
9. ABC is an equilateral triangle and ACDE is a square drawn outside it. Find $\angle ABE$.
10. In triangle ABC, $AB = AC$ and D is the mid-point of BC. Prove that AD is perpendicular to BC.

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Answer Key

1. 76°
2. 40°
3. $\angle Q = 50^\circ = \angle P$, so $QR = PR$
4. $50^\circ, 65^\circ, 65^\circ$
5. 80°
6. 8 cm
7. 32°
8. Equilateral
9. 15°
10. $\triangle ABD \cong \triangle ACD$ (S.S.S.), so $\angle ADB = \angle ADC$; they add up to 180°, so each is 90°

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